

Highlights

State of the art in energy communities in Ibero-America: technology, governance, and regulation for the social appropriation of energy

RIPCEL Research Consortium

- PRISMA review of 55 empirical studies on ECs in 10 countries.
- Four-dimensional taxonomy for an Ibero-American EC Observatory.
- Regulatory divide between Iberian Peninsula and Latin America.
- Cooperative models prevail with hybrid governance innovations.
- Nine evidence-based policy recommendations for EC deployment.

State of the art in energy communities in Ibero-America: technology, governance, and regulation for the social appropriation of energy

RIPCEL Research Consortium^a

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Abstract

Energy communities (ECs) represent an emerging paradigm for just energy transitions in Ibero-America, combining renewable technologies with participatory governance to address energy poverty and strengthen territorial resilience. However, empirical evidence remains fragmented across linguistic, disciplinary, and geographic boundaries. This article presents a systematic review following the PRISMA 2020 protocol, synthesizing 55 empirical studies from 10 Ibero-American countries. The review is structured around four analytical dimensions—technology, governance, regulation, and social appropriation—forming the foundational taxonomy for a proposed Ibero-American Observatory of Energy Communities. In the reviewed sample, key findings reveal: (i) predominance of solar photovoltaic technology (89% of cases) with growing microgrid and storage integration; (ii) prevalence of cooperative models (64%) with hybrid innovations combining formal and informal governance, including indigenous self-governance; (iii) a regulatory divide between enabling Iberian frameworks (supported by EU directives) and fragmented Latin American landscapes; and (iv) predominant orientation toward energy justice (67%), emphasizing poverty reduction in rural, indigenous, and marginalized urban communities. Six critical knowledge gaps are identified, and an open-repository architecture for systematizing regional experiences is proposed. Policy recommendations include adopting EC-specific legislative frameworks in Latin America, developing financing mechanisms for low-income communities, and advancing regional regulatory harmonization.

Keywords: Energy communities, Participatory governance, Energy transition, Ibero-America, Social appropriation, Energy observatory

Nomenclature

EC	Energy community/communities
PV	Photovoltaic
DG	Distributed generation
P2P	Peer-to-peer (energy trading)
AMI	Advanced metering infrastructure
MAMCA	Multi-actor multi-criteria analysis
RED II	EU Renewable Energy Directive (recast)
IEMD	EU Internal Electricity Market Directive
RUCE	Single Registry of Energy Communities (Colombia)
MLP	Multi-Level Perspective
TIS	Technological Innovation Systems
SNM	Strategic Niche Management
SLR	Systematic literature review
MMAT	Mixed Methods Appraisal Tool

1. Introduction

The global energy transition demands innovative approaches that go beyond centralized energy systems and empower local communities as active agents in the generation, distribution, and governance of renewable energy. Energy communities (ECs), defined as collective initiatives in which citizens, local authorities, and small enterprises collaborate to produce, consume, and manage renewable energy, represent a paradigm shift toward decentralized, democratic, and socially just energy systems ([Koirala et al., 2016](#); [Bauwens et al., 2016](#); [Walker and Devine-Wright, 2008](#)).

The Ibero-American space, encompassing the Iberian Peninsula (Spain and Portugal) and thirteen Latin American countries, faces unique energy challenges and opportunities. While the region has one of the cleanest electricity matrices in the world, with a renewable share fluctuating between 59% and 61% of total generation ([OLADE, 2023](#)), critical asymmetries persist in access, affordability, and climate resilience. Regional energy intensity has improved by an average of only 1% per year over the past decade, falling short of the 2% global average required to meet the Paris Agreement targets ([CEPAL, 2024](#)). Moreover, renewable generation losses reached 53,000 GWh in 2024 due to system flexibility constraints ([World Economic Forum, 2025](#)).

In this context, the emergence of ECs responds not only to a technical imperative for decarbonization but also to a social justice imperative aiming to transform end-users from passive consumers into active prosumers under collective ownership and participatory governance schemes ([Szulecki, 2018](#); [Jenkins et al., 2016](#)). The Iberian Peninsula benefits from EU regulatory frameworks that promote ECs, whereas Latin America faces persistent energy poverty, regulatory fragmentation, and grassroots innovations with limited institutional capacity ([Canavire-Bacarreza and Ponce-Rodriguez, 2022](#); [Guzowski and Recalde, 2021](#)).

Despite growing interest, systematic evidence synthesis on ECs in Ibero-America is limited. Existing reviews focus on Northern European contexts or individual countries, lacking comprehensive analyses spanning the techno-

logical, governance, regulatory, and social justice dimensions that characterize Ibero-American ECs (Hanke et al., 2021; Lowitzsch, 2020). This fragmentation is exacerbated by linguistic barriers, the distribution of relevant literature across Spanish, Portuguese, and English, and the disconnect among academia, public policy, and community practice.

Recent reviews have begun to address these gaps. Peeters et al. (2025) Identify critical gaps in EC understanding and assessment, warning of an optimistic bias in the literature that tends to overestimate benefits without rigorous impact evaluations. Polanco and García-Martínez (2025) applies the multi-level perspective to grid-connected Latin American ECs. Nevertheless, no prior review has systematically integrated the four dimensions, technology, governance, regulation, and social appropriation, in a multilingual analysis spanning both the Iberian Peninsula and Latin America.

This article addresses this gap through a systematic literature review, conceived as original empirical research employing SLR methodology to generate novel findings, synthesizing 55 empirical studies from 10 Ibero-American countries, organized around four analytical dimensions: (i) appropriate technologies, (ii) governance models, (iii) regulatory and financial frameworks, and (iv) social appropriation and energy justice. These dimensions constitute the foundational taxonomy for the construction of an Ibero-American Observatory of Energy Communities, which, as an open repository, would enable systematic comparison and scaling of regional experiences.

The specific objectives of this article are:

- (i) Characterize the state of the art in EC research and practice in Ibero-America, identifying geographic, temporal, and thematic patterns.
- (ii) Systematize the technologies, governance models, regulatory frameworks, and social appropriation processes documented in the empirical literature.
- (iii) Identify critical knowledge gaps that define the future research agenda.
- (iv) Propose a taxonomic and operational architecture for an Ibero-American EC Observatory functioning as an open knowledge infrastructure.
- (v) Derive evidence-based policy recommendations for the design of enabling frameworks in the region.

1.1. Theoretical framework

This article draws on three complementary theoretical frameworks, supplemented by two additional analytical tools, that inform the organization of the analysis. The combination of MLP, TIS, and commons governance follows the integrative approach advocated by Markard et al. (2012) for sustainability transitions research.

Multi-Level Perspective (MLP). The socio-technical transitions theory of Geels and Schot (2007) provides the framework for understanding how ECs emerge as innovation niches that challenge the dominant centralized energy regime, conditioned by the macroeconomic and regulatory landscape. In Ibero-America, the regime level comprises vertically integrated state-owned utilities (e.g., CFE in Mexico, Ecopetrol/ENEL in Colombia, Eletrobras in Brazil, Endesa in Chile) and centralized market designs that structurally resist distributed community ownership. At the landscape level, critical pressures include the Paris Agreement and Nationally Determined Contributions; volatile

fossil fuel commodity prices (particularly natural gas dependency in Argentina and Chile); and political cycles that alternately open and close policy windows, such as Colombia’s 2022 progressive government establishing the first EC decree, Mexico’s 2018, 2024 re-centralization toward CFE, and Brazil’s post-2022 regulatory uncertainty following Law 14.300. This perspective is particularly relevant for explaining the divergent trajectories between the Iberian Peninsula, where the European regulatory regime catalyzes the transition through a *reconfiguration pathway* (Geels and Schot, 2007) in which incumbent utilities accommodate niche innovations, and Latin America, where ECs operate as niches protected by international cooperation and academic networks in a phase of *strategic niche management* (SNM) (Schot and Geels, 2008; Raven et al., 2010), sustained through shielding (international funding), nurturing (university-community partnerships), and early empowerment (pilot-to-policy advocacy).

Technological Innovation Systems (TIS). While MLP illuminates the multi-level dynamics of EC transitions, the TIS framework (Bergek et al., 2008; Hekkert et al., 2007) provides complementary analytical tools for assessing the functional performance of EC innovation systems. We employ five TIS functions as diagnostic categories in the Discussion: (F1) knowledge development and diffusion, (F2) resource mobilization, (F3) market formation, (F4) legitimacy creation, and (F5) entrepreneurial experimentation. This combined MLP+TIS approach follows Markard et al. (2012) and enables analysis of both the structural position of ECs within the socio-technical system and the functional dynamics that accelerate or impede their development.

Governance of the Commons. The institutional framework of Ostrom (2009) for common-pool resource management grounds the governance dimension analysis, recognizing that distributed renewable energy can function as a common good managed through polycentric community arrangements. Ostrom’s eight design principles, defined membership (DP1), locally adapted rules (DP2), collective decision-making (DP3), monitoring (DP4), graduated sanctions (DP5), conflict resolution (DP6), recognized autonomy (DP7), and nested organization (DP8), provide analytical criteria for evaluating the robustness of documented governance structures. We operationalize these principles in Section 3.3 by assessing their presence or absence across the documented governance cases.

Energy justice. The energy justice framework, originally proposed as a triad of distributive, procedural, and recognition justice by Jenkins et al. (2016) and developed by Sovacool and Dworkin (2014), structures the social appropriation dimension analysis. We adopt the expanded four-dimensional conceptualization that incorporates *restorative justice* (Heffron and McCauley, 2017), recognizing the need to remedy historical harms caused by extractive energy systems in Indigenous and Afro-descendant territories. Following the comprehensive methodological review by Jenkins et al. (2021), we also attend to the cosmopolitan dimensions of energy justice that transcend national boundaries, particularly relevant in the cross-border Ibero-American context. In the Ibero-American setting, this framework is enriched by the concept of “social appropriation of science, technology, and innovation” (*apropiación social de la ciencia, la tecnología y la innovación*) from the Latin American tradition of social studies of science (Szulecki, 2018), which emphasizes community agency in the co-construction of technical systems.

These frameworks converge in the proposed four-dimensional taxonomy. The technological dimension captures niche configuration (MLP) and innovation functions (TIS). The governance dimension (analyzed through Ostrom’s principles) provides the instrumental framework, operational architecture and institutional rules required to manage energy as a common good. The regulatory dimension reflects regime and landscape conditions (MLP). Finally, the social appropriation dimension acts as the axiological framework, which defines the core ethical values through the

four pillars of energy justice and emphasizes community agency in the co-construction and dignified appropriation of technical systems.

2. Methodology

2.1. Protocol and registration

This systematic review follows the PRISMA 2020 guidelines (Page et al., 2021). The protocol was developed in March 2026 within the framework of the Ibero-American RIPCEL Network, funded by the CYTED program. The prospective registration would have been preferable; the retrospective registration includes the complete search strategy, eligibility criteria, and analysis plan, all of which were finalized prior to data extraction.

2.2. Search strategy

Exhaustive searches were conducted across four sources: Scopus, Web of Science (WoS Core Collection), SciELO, and RedALyC, using 17 multilingual queries (in Spanish, Portuguese, and English) and Boolean operators, targeting 15 Ibero-American countries. The combination of Scopus and WoS ensures coverage of high-impact, indexed international literature, while SciELO and RedALyC were specifically incorporated to capture Latin American literature in Spanish and Portuguese that may be underrepresented in mainstream international databases. To cast a wider net and capture grey literature, supplemental backward and forward snowballing was performed.

Search terms spanned six thematic dimensions: renewable technologies, governance models, regulatory frameworks, institutional actors, dissemination strategies, and social appropriation of energy. Geographic terms explicitly included all 15 target countries, as well as regional descriptors ("Ibero-America," "Latin America").

Time range: 2010–2025. Languages: English, Spanish, and Portuguese. Initial results: 1,154 articles identified. All records were managed using Zotero reference management software for manual deduplication and screening stages. An example of the search logic applied in Scopus and WoS is: TITLE-ABS-KEY ("energy community" OR "comunidad energética" OR "comunidade de energia") AND ("Ibero-America" OR "Latin America" OR "Spain" OR "Portugal" OR [Country List...]). The full list of search strings and the search protocol are available in the OSF repository.

2.3. Eligibility criteria

Eight screening criteria were developed (five inclusion, three exclusion):

Inclusion: (1) geographic focus on at least one of the 15 RIPCEL countries; (2) empirical content (case studies, field data, pilot projects); (3) coverage of at least one of the six thematic dimensions; (4) inclusion of community governance or social impact dimensions; (5) novel approach applicable to the Ibero-American context.

Exclusion: (6) purely theoretical or simulation studies without empirical basis; (7) contexts outside Ibero-America without comparative analysis; (8) large-scale centralized systems without community governance.

2.4. Selection process

Stage 1 – Deduplication: 232 duplicates removed manually using Zotero’s duplicate detection and subsequent human verification; 922 unique articles retained.

Stage 2 – Title and abstract screening: Title and abstract reading and evaluation were performed in a human-AI collaborative framework. A human reviewer and an AI assistant (Claude, in its co-work version) independently processed the titles and abstracts. In cases of discrepancy, the human reviewer performed an additional screening of the article’s conclusions to make the final inclusion decision. In this stage, 862 articles were excluded (primarily for non-Ibero-American context, being purely theoretical, focusing on centralized systems, or lacking community governance). The remaining 60 articles advanced to full-text screening.

Stage 3 – Full-text screening: Complete articles were retrieved and assessed against the eligibility criteria by two independent human reviewers, with discrepancies resolved by consensus. Five articles were excluded (3 for insufficient empirical detail, 2 for lack of novelty). Thus, 55 studies were included in the final qualitative synthesis (Figure 1).

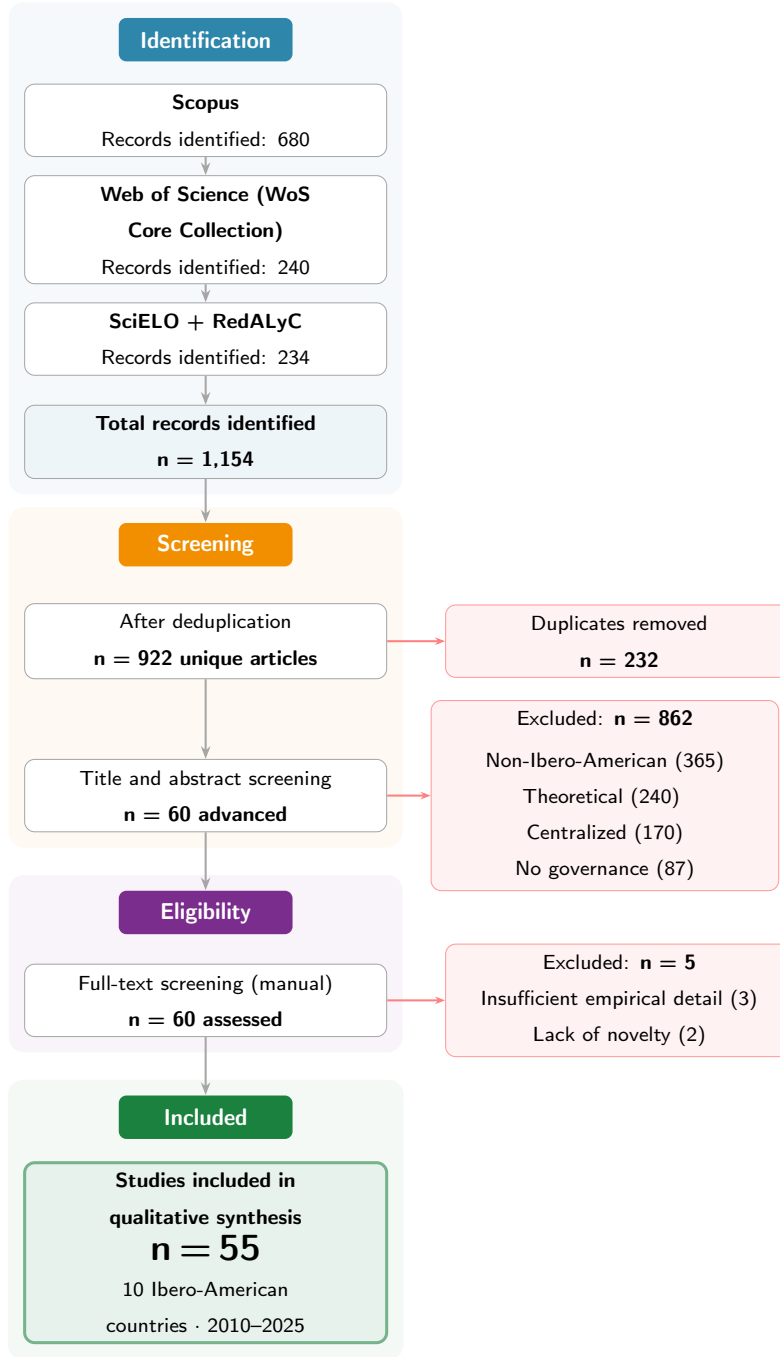


Figure 1: PRISMA 2020 flow diagram of the study selection process. The identification (blue), screening (orange), eligibility (purple), and inclusion (green) phases are shown with their respective exclusion criteria (red).

2.5. Data extraction and synthesis

Structured data were manually extracted from each study for the six thematic dimensions. A standardized coding framework was utilized to perform (i) entity extraction (technology types, governance structures, country, and legal frameworks); (ii) thematic classification (mapping content to the six dimensions); and (iii) summarization of findings per dimension. Extraction was performed independently by two human reviewers for the full set of 55 studies, with a third reviewer resolving discrepancies to ensure consistency. Overall inter-coder reliability (Cohen’s κ)

was calculated for the entire set, resulting in $\kappa = 0.85$; broken down by dimension: technology $\kappa = 0.92$ (high, due to objective categorization); governance $\kappa = 0.81$, regulation $\kappa = 0.88$, and social appropriation $\kappa = 0.78$ (lowest, reflecting the inherent subjectivity of justice categorization). Within the social appropriation dimension, disagreements concentrated on: (a) distinguishing procedural from recognition justice (accounting for 60% of coding discrepancies), and (b) attributing restorative justice intent to studies that document historical remediation without employing the term explicitly. The restorative justice findings in Section 3 should therefore be interpreted as a lower-confidence interpretive category. Given the heterogeneity of methodological designs (case studies 51%, comparative studies 27%, surveys 22%), narrative synthesis was employed following the SWiM (*Synthesis Without Meta-analysis*) guidelines (Campbell et al., 2020). Specifically, the SWiM approach was implemented through: (a) grouping studies by thematic dimension and sub-regional context (Iberian vs. Latin American); (b) reporting vote counting as a descriptive approach (frequencies and proportions) rather than as an effect size estimator; (c) planned contrasts comparing governance models, regulatory environments, and justice outcomes between the two sub-regions; and (d) examining how study design heterogeneity might affect the synthesis, with single-case studies providing depth and comparative/survey studies providing breadth.

2.6. Quality assessment

The methodological quality of the 55 included studies was assessed using the *Mixed Methods Appraisal Tool* (MMAT, version 2018), which was applied independently by two researchers, with a third resolving disagreements. Results indicate that 62% of studies (n=34) received a high quality rating (≥ 4 criteria satisfied), 29% (n=16) medium quality (3 criteria), and 9% (n=5) low quality (≤ 2 criteria). Low-quality studies are mainly conference papers and technical reports that lack sufficient methodological detail. They were included in the synthesis to preserve geographic breadth, but their findings are flagged with caution markers [LQ] in the analysis. Individual MMAT scores for all 55 studies are provided in Supplementary Table S1.

2.7. Search for laws, regulations, and policy documents

A key limitation of conventional bibliographic databases (Scopus, WoS, SciELO) is their near-exclusion of primary legal instruments, government decrees, regulatory resolutions, and institutional technical reports—the very documents that govern energy communities in practice. To close this gap, the methodology incorporates a complementary, active search in official sources, structured around three pillars:

Pillar 1 — Systematic search in official records. Because there is no consolidated database for Ibero-American energy legislation, a country-by-country active search was conducted for: (a) national laws, decrees, resolutions, and regulatory rulings published in official gazettes; (b) technical documents and reports issued by ministries of energy, utility regulators, development banks, and intergovernmental organizations (OLADE, CEPAL/ECLAC, IDB, IDAE); and (c) grey literature from NGOs and think tanks specialized in energy policy. Searches used multilingual Boolean strings designed to capture both the formal nomenclature of energy community legislation and indirect references to community-scale generation projects (e.g., "comunidad energética" OR "autoconsumo colectivo" OR "generación distribuida comunitaria" combined with country and date filters).

Pillar 2 — Specialized policy databases. Instead of conventional academic repositories, the search drew on institutional observatories and legal databases: the IEA Policy Database for comparative regulatory data; the

CEPAL/ECLAC Energy Efficiency Information Base (BIEE) for regional policy instruments; the OLADE Energy Information System (SIEE) for national statistics; the Spanish IDAE Observatory for Autonomous Community norms; and official regulatory repositories in Colombia (CREG), Mexico (SENER/CENACE), Brazil (ANEEL), Chile (CNE), and Portugal (Ministry of Energy Transition). These sources enabled systematic classification of regulatory instruments into three functional categories: (i) legislation and binding regulation; (ii) energy planning instruments; and (iii) financing and incentive mechanisms.

Pillar 3 — Interpretive triangulation. Once the normative corpus was compiled, an interpretive analysis identified regulatory patterns, cross-country divergences, institutional gaps, and structural barriers. A distinctive feature of this approach is the triangulation of legal-institutional intelligence with the empirical findings from the scientific literature reviewed in Sections 2.2–2.5. The inclusion of these primary regulatory documents, which are not available in purely academic articles, enables the analysis to situate theoretical trends within the specific sociopolitical and territorial realities of each country, yielding the contextual granularity required for actionable policy recommendations.

3. Results

3.1. Study characteristics

The 55 included studies show an uneven geographic distribution: Spain (n=18, 33%), Colombia (n=8, 15%), Brazil (n=7, 13%), Portugal (n=6, 11%), Mexico (n=4, 7%), Chile (n=3, 5%), Argentina (n=3, 5%), Peru (n=2, 4%), Costa Rica (n=1, 2%), and multi-country analyses (n=3, 5%). No studies were identified from Cuba, Ecuador, Honduras, Panama, Paraguay, or the Dominican Republic (Table 1).

Eighty percent of studies were published between 2019 and 2025, indicating an emerging and rapidly expanding research field. The predominant methodological designs were individual case studies (51%), multiple comparative studies (27%), and empirical surveys/interviews (22%). The corpus of N=55 studies is comparable to other recent systematic reviews in this field: [Peeters et al. \(2025\)](#) reviews 62 studies on EC assessment frameworks, while [Polanco and García-Martínez \(2025\)](#) analyzes 38 Latin American cases. Given this sample size, all proportions reported hereafter should be interpreted as descriptive frequencies of the reviewed corpus rather than as population-level estimates.

Table 1: Geographic and temporal distribution of included studies (N=55).

Country/Region	n	%	2010–2018	2019–2025
Spain	18	33	6	12
Colombia	8	15	1	7
Brazil	7	13	2	5
Portugal	6	11	1	5
Mexico	4	7	2	2
Chile	3	5	1	2
Argentina	3	5	0	3
Peru	2	4	0	2
Costa Rica	1	2	1	0
Multi-country	3	5	0	3
Total	55	100	14	41

3.2. Dimension 1: Appropriate technologies for energy communities

In the reviewed sample, solar photovoltaic (PV) systems are the most frequently documented technology in Ibero-American ECs, present in 49 of 55 studies (89%), followed by microgrids (31%), hybrid systems (22%), energy storage (18%), smart grids (16%), wind (15%), hydrokinetic (9%), and biomass (5%) (Table 2).

Table 2: Technologies documented in Ibero-American energy communities (N=55).

Technology	n	%	Typical scale / Contexts
Solar photovoltaic	49	89	5 kW (household) to 5 MW (municipal); urban and rural
Microgrids	17	31	Off-grid communities; islands; resilience
Storage (batteries)	10	18	10–500 kWh; lithium-ion and lead-acid
Smart grids	9	16	Advanced metering; P2P; demand management
Wind	8	15	<100 kW (micro-wind); coastal/highland
Hybrid systems	12	22	Solar+wind; solar+diesel (transitional)
Hydrokinetic	5	9	Micro-turbines; Andean and Central American regions
Biomass	3	5	Agricultural waste; biogas

Key findings from the reviewed corpus:

Photovoltaic prevalence. The high frequency of solar PV in the reviewed literature reflects technological maturity, cost reductions (80% over the 2010–2020 decade), modularity for community-scale applications, and abundant solar resources across the region. Configurations range from rooftop installations for collective self-consumption in Spain (Molina et al., 2022; López et al., 2022b) to solar microgrids with storage for rural hospitals in Colombia (Istvan et al., 2024) and solar cooperatives in Brazilian favelas (Campos et al., 2021). However, this prevalence may partly reflect a publication bias, as PV projects are more visible and documented, while biomass or small hydropower experiences in rural communities may be underrepresented in indexed academic literature (Peeters et al., 2025).

Microgrids and storage. Microgrids emerge in the literature as critical infrastructure for communities without grid access or with low reliability (Kirubi, 2020; Santos et al., 2023). The Vichada Hospital experience (Colombia) documents a 50 kW PV microgrid with 100 kWh storage that replaced diesel generators, reducing energy costs by 70% (Istvan et al., 2024). However, storage represents 30–50% of total system cost, constituting a significant barrier (Martínez et al., 2025).

Digital frontier. Emerging experiences of blockchain-based P2P trading (Colombia, La Estrecha Solar) and smart management platforms are documented (López et al., 2022a). Nevertheless, the digital divide limits applicability in rural and low-income communities: while Uruguay achieves 100% AMI coverage (OLADE, 2023), Colombia registers only 3% deployment (Martínez-Rengifo and Fernández-Correa, 2025).

Technology appropriation. Several studies emphasize the importance of participatory technology selection, integrating community priorities (health, education, productive activities) with technical feasibility (Mundaca et al., 2018; Nino and Swain, 2021). Documented conflicts in Chile over wind projects imposed without consultation underscore that “dignified energy appropriation” (*apropiación digna de la energía*, a concept from the Latin American tradition emphasizing community agency in technology co-design) requires community participation from the design stage (Fuentes Gonzalez et al., 2019; Argote-Cusi, 2017).

3.3. Dimension 2: Governance models and social participation

In the reviewed sample, cooperative models are the most frequently documented governance structure in Ibero-American ECs (35 of 55 studies, 64%), followed by community associations (27%), municipal/public entities (15%), informal structures (11%), and ethnic self-governance (7%) (Table 3).

Table 3: Governance models in Ibero-American energy communities (N=55).

Model	n	%	Mechanism and example
Cooperative	35	64	One member, one vote; Som Energia (Spain, 86,000 members)
Community association	15	27	Community action boards (Colombia); neighborhood associations
Municipal/public entity	8	15	Public-community partnerships; municipal management
Informal structure	6	11	Unregistered assemblies; grassroots collectives
Ethnic self-governance	4	7	Indigenous autonomy; councils of elders (Mexico)

Key findings from the reviewed sample:

Iberian cooperativism. Spanish and Portuguese cooperatives follow formalized legal frameworks under the RED II Directive, with mature structures of general assemblies, governing boards, and specialized committees (Capellán-Pérez et al., 2018; Díaz et al., 2023; Sagebiel, 2019). Som Energia (Catalonia, 86,000 members), GoiEner (Basque Country, 17,611 members), and Grupo Enercoop (Alicante) exemplify model consolidation, with 44% year-on-year growth in Spain, reaching 659 ECs by the end of 2024.

Latin American hybrid governance. Latin America exhibits more diverse and hybrid governance structures (Martínez-Rengifo and Fernández-Correa, 2025; Campos et al., 2021; Polanco and García-Martínez, 2025). In Colombia, community action boards partner with universities and NGOs to implement solar projects, integrating formal boards with community assemblies. In Brazil, favela cooperatives combine cooperative legal structures with sustainability committees and monthly participatory meetings. In Mexico, indigenous communities integrate energy decisions into their traditional governance systems (community assemblies, councils of elders). These hybrid arrangements can be analyzed as polycentric governance systems in the sense of Ostrom (2009), in which multiple decision centers operate with local autonomy and regional coordination.

Ostrom’s design principles assessment. Table 4 presents a cross-tabulation of Ostrom’s eight design principles against the principal governance archetypes documented in the reviewed literature. This assessment is based on the coded evidence: “✓” indicates the principle is explicitly documented as present, “~” indicates partial or emerging compliance, and “—” indicates absence or insufficient evidence.

Table 4: Assessment of Ostrom’s design principles across governance archetypes.

DP	Principle	Iberian Coop.	LA Assoc.	Favela Coop.	Indigenous
1	Defined membership boundaries	✓	✓	~	✓
2	Locally adapted rules	✓	✓	✓	✓
3	Collective decision-making	✓	✓	✓	✓
4	Monitoring	✓	~	—	~
5	Graduated sanctions	✓	—	—	✓
6	Conflict resolution	✓	~	~	✓
7	Recognized autonomy	✓	~	—	✓
8	Nested organization	✓	—	—	~

Iberian cooperatives satisfy all eight principles, reflecting their institutional maturity. Indigenous ECs exhibit strong compliance with principles rooted in traditional governance (DP1–3, DP5–7) but weaker compliance with monitoring (DP4) and nested organization (DP8), the latter requiring inter-community coordination mechanisms that remain underdeveloped. Latin American associations and favela cooperatives show consistent compliance with participatory principles (DP2–3) but gaps in monitoring, sanctions, and external recognition—patterns that correlate with the challenges documented in the literature: participation fatigue, elite capture, and institutional fragility (Díaz et al., 2023; Jorge and Ortega Arango, 2023; Vargas et al., 2022).

Participatory mechanisms. The most effective mechanisms documented include: regular general assemblies, specialized working groups, consensus-building processes, and digital platforms for information and decision-making (Lode et al., 2023, 2022). The Vega de Valcarce EC (Spain) employed a MAMCA involving residents, municipal government, university, and private partners in project co-design (Lode et al., 2023).

Critical success factors and survivorship bias. ECs with robust participatory governance report higher member satisfaction, greater willingness to invest in expansion, and stronger social cohesion (Campos et al., 2021; Lode et al., 2022). However, challenges persist: participation fatigue, power imbalances, elite capture, regulatory complexity, and financial sustainability (Díaz et al., 2023; Jorge and Ortega Arango, 2023; Vargas et al., 2022). Following the warning of Peeters et al. (2025), it must be noted that the reviewed studies overwhelmingly document ECs in consolidation phases: none reports a failed or abandoned project, which may bias the perception of success factors and constitutes a significant survivorship bias in the available literature.

3.4. Dimension 3: Regulatory and financial frameworks

The regulatory landscape reveals a structural divide between the Iberian Peninsula and Latin America (Table 5).

Table 5: Comparative regulatory frameworks for energy communities in Ibero-America.

Country	Enabling framework	Key provisions	Main barriers	Status	Ref.
Spain	RD 244/2019; RDL 23/2020	Legal recognition of ECs; collective self-consumption; tax exemption	Incumbent resistance; administrative complexity	Active	IDAE 2024
Portugal	DL 162/2019; DL 15/2022	REC and CEC; collective self-consumption; municipal support	Recent implementation; limited grid capacity	Active	VDA 2022
Colombia	Law 1715/2014; Res. CREG 101,072/2025; Decree 2236/2023	Tax incentives; DG <5 MW; RUCE registry	No specific EC framework; implementation gap	Active	CREG 2025
Brazil	Res. ANEEL 482/2012; Law 14.300/2022	Net metering; shared generation; cooperatives	No EC recognition; recent regulatory uncertainty	Amended	ANEEL 2022
Mexico	Energy Transition Law (2015); 2025 Reform	Net metering; DG limit 0.7 MW; energy justice	Regulatory instability; CFE centralization	Amended*	CEMDA 2025
Chile	Law 21.118 (Net Billing)	Generation ≤ 300 kW; collective grouping	No legal EC definition; concentrated market	Active	Biblioteca del Congreso Nacional de Chile 2018
Argentina	Law 27.191; provincial laws	Provincial net metering; virtual community DG ≤ 12 MW	Provincial fragmentation; economic instability	Active	Ibarra and Chiabrando 2022

*Mexico's Energy Transition Law was undermined by 2021–2024 constitutional reforms before the 2025 reform.

Key findings from the reviewed sample:

Iberian enabling frameworks. Spain and Portugal benefit from transposition of the European RED II and IEMD directives, providing legal recognition, simplified procedures, and political support for ECs ([Capellán-Pérez et al., 2018](#); [Neves and Torstensson, 2024](#); [Santos et al., 2023](#)). Spain leads with innovative autonomous community legislation: the Basque Country (Law 1/2024) guarantees a minimum 20% ownership reserve in new parks for local members; the Balearic Islands (Law 5/2024) regulates closed networks and territorial solidarity funds; and Navarra (Foral Law 4/2025) sets a target of 70 active ECs by 2030.

Latin American fragmentation. Most Latin American countries have adopted distributed generation and net-metering frameworks focused on individual installations rather than on collective community arrangements ([Poque Gon-](#)

zalez et al., 2023; Campos et al., 2021; Leonarda et al., 2022). Brazil leads with its “shared generation” modality but lacks the governance and social dimensions emphasized in European EC definitions. Colombia advances with Decree 2236/2023, which distinguishes self-generation ECs from commercialization ECs, and Resolution CREG 101,072/2025, which enables energy compensation at the injection point against individual member consumption.

Financing mechanisms. Documented instruments include: public funds (IDAE in Spain, FENOGE in Colombia, FAE in Chile), international cooperation (IDB, World Bank, GIZ), social economy (cooperative capital, crowdfunding), and innovative mechanisms (Tax Credit Works in Colombia, Green Bonds from CoopeGuanacaste in Costa Rica). However, access to financing remains the most critical barrier for low-income communities (CEPAL, 2024; Díaz et al., 2023). High-quality studies in our sample ($n=34$, $\text{MMAT} \geq 4$) report financing gaps of USD 500–3,000 per capita for 5–50 kW systems, a prohibitive range without direct subsidies for communities below the multidimensional poverty line.

3.5. Dimension 4: Social appropriation and energy justice

Social appropriation of energy constitutes both a central motivation and a verifiable outcome of Ibero-American ECs in the reviewed literature. In the sample, 37 of 55 studies (67%) explicitly address energy justice, energy poverty, energy democracy, or community empowerment.

Distributive justice. ECs address distributive injustices by providing affordable access to electricity for marginalized populations. The Vichada Hospital microgrid (Colombia) provides 24/7 electricity to an Amazonian indigenous community, improving healthcare access (Istvan et al., 2024; Nino and Swain, 2021). The Percila e Lúcio Cooperative in a Rio de Janeiro favela reduces electricity costs by 30–60%, alleviating urban energy poverty (Campos et al., 2021). In the Colombian Chocó, 23 Educational Energy Communities inaugurated in 2024 provide permanent solar energy to educational institutions in remote areas, benefiting over 1,400 students.

Procedural justice. ECs advance procedural justice through participatory governance, democratic control, and community empowerment (Lode et al., 2023; Jorge and Ortega Arango, 2023). However, the Isthmus of Tehuantepec experience in Mexico illustrates tensions: large-scale wind farms have excluded local communities from decision-making, generating opposition (Argote-Cusi, 2017; Ramirez, 2020). The challenge is moving from “formal consultation” to “substantive participation” (Baker, 2016).

Recognition justice. Recognition of local knowledge, cultural practices, and territorial rights of indigenous peoples is a distinctive dimension of the Ibero-American context (Baker, 2016; Mundaca et al., 2018; Vargas et al., 2022). Colombia’s ethnic differential approach model allows indigenous, Black, and Afro-Colombian communities to constitute ECs under their own self-governance norms.

Restorative justice. Following the expanded energy justice framework of Heffron and McCauley (2017), several documented ECs address historical harms caused by extractive energy systems. In the Isthmus of Tehuantepec, community solar projects explicitly aim to restore territorial agency undermined by decades of corporate wind energy extraction without local benefit (Argote-Cusi, 2017). In Colombia’s Chocó, educational ECs serve as partial remedy for centuries of energy exclusion in Afro-descendant territories. While the reviewed literature does not explicitly employ the term “restorative justice”, the underlying logic, remedying historical energy injustice through community-controlled alternatives, is present in at least 11 studies (20%).

Gender equity. Gender equity remains underexplored: only 9% of studies explicitly address gender dimensions. Identified gaps include female underrepresentation in membership and leadership roles, male dominance in executive positions, and differential impacts of the energy transition on women (Vargas et al., 2022).

Climate resilience. Community microgrids and storage systems enable the maintenance of essential services during extreme weather events. The post-Hurricane María experience in Puerto Rico demonstrated that community solar microgrids, combined with local self-management, constitute the most resilient disaster response (Kirubi, 2020).

4. Discussion

4.1. Comparative synthesis through an MLP+TIS lens

Results reveal distinctive patterns between the two sub-continentals comprising the Ibero-American space (Table 6), which can be systematically analyzed through the combined MLP and TIS frameworks.

Table 6: Comparative analysis between the Iberian Peninsula and Latin America.

Dimension	Iberian Peninsula	Latin America
Regulation	Harmonized frameworks (EU RED II); legal recognition	Fragmentation; absence of EC-specific frameworks
Governance	Formalized cooperatives; professional management	Hybrid governance; indigenous and grassroots innovation
Technology	Grid-connected; collective self-consumption	Off-grid microgrids; hybrid systems
Financing	EU funds; mature cooperative capital	Limited public funds; international cooperation
Social justice	Energy democracy; tariff advocacy	Energy poverty; indigenous sovereignty; universal access
Key actors	Cooperatives + municipalities	Universities + NGOs + communities

MLP transition pathways. The Iberian Peninsula follows what Geels and Schot (2007) characterize as a *reconfiguration pathway*: incumbent utilities (Endesa, EDP, Iberdrola) accommodate niche innovations through regulatory adaptation, incorporating community energy options into their portfolios while maintaining dominant market positions. The regulatory regime has been partially reconfigured—not substituted—to create space for ECs. In contrast, Latin American ECs operate predominantly within a *de-alignment/re-alignment* dynamic: the erosion of centralized regime legitimacy (e.g., blackouts in Colombia, energy poverty in Mexico, tariff crises in Argentina) creates space for niche experimentation, but no coherent realignment has occurred at the regime level. Mexico represents a counterfactual: the 2018–2024 re-centralization toward CFE actively closed policy windows for niche upscaling, illustrating how landscape pressures (political nationalism, state-enterprise ideology) can reinforce rather than destabilize incumbent regimes.

TIS functional analysis. Applying TIS diagnostic categories (Bergek et al., 2008; Hekkert et al., 2007) to our findings reveals asymmetric functional performance:

- **F1 – Knowledge development and diffusion:** Strong in both sub-regions but linguistically fragmented. The SciELO/RedALyC finding of zero additional unique articles suggests that Spanish/Portuguese regional literature is either captured by other databases or remains in grey literature. Knowledge exchange between the Iberian Peninsula and Latin America is limited, representing a systemic bottleneck.
- **F2 – Resource mobilization:** Asymmetric. Iberian ECs access EU structural funds and mature cooperative capital; Latin American ECs rely on volatile international cooperation and project-based university funding, posing a sustainability risk to the innovation system.
- **F3 – Market formation:** Nascent in Latin America (no EC-specific tariff regimes), mature in Iberia (collective self-consumption, virtual net metering).
- **F4 – Legitimacy creation:** Iberian ECs benefit from EU-level political endorsement; Latin American ECs build legitimacy through grassroots social movements and indigenous sovereignty claims—a distinct legitimization pathway with different political risks.
- **F5 – Entrepreneurial experimentation:** Paradoxically stronger in Latin America, where regulatory gaps enable informal experimentation (favela cooperatives, indigenous microgrids) that would not be permitted under Iberian regulatory frameworks.

A future research direction—beyond the scope of this review—would be to connect TIS functions with measurable indicators (e.g., patent counts and publications for F1, investment flows for F2, installed capacity and tariff regimes for F3) to enable quantitative benchmarking of EC innovation systems across countries.

4.2. Critical knowledge gaps

The analysis identifies six gaps defining the future research agenda. These gaps converge with those identified by [Peeters et al. \(2025\)](#), who warn that EC research globally suffers from an optimistic bias leading to overestimation of benefits and underestimation of barriers, failures, and negative impacts. Our review confirms this pattern in the Ibero-American context: of the 55 included studies, none explicitly reports a failed or abandoned EC project, and only 18% (n=10) document barriers in detail.

1. Gender equity: Only 9% of studies explicitly address gender dimensions in ECs. Specific research on female participation, leadership, and differentiated impacts of the energy transition is needed.
2. Unrepresented countries: No studies were identified from Cuba, Ecuador, Honduras, Panama, Paraguay, or the Dominican Republic, revealing significant gaps in Central America and the Caribbean.
3. Longitudinal studies: Most research is cross-sectional. Studies tracking EC evolution over time are needed to assess long-term sustainability, including data on ECs that ceased operations.
4. Impact assessment: Rigorous evaluations of EC impacts on energy poverty, well-being, local economic development, and carbon emissions are scarce. [Peeters et al. \(2025\)](#) emphasizes that the absence of standardized multidimensional performance assessment frameworks is one of the field’s most critical gaps.

5. Open science: Only 11% of studies adopt open science practices. The region needs open repositories, shared data, and participatory research.
6. Standardized taxonomy: No consensus classification of ECs in Ibero-America exists. Definition heterogeneity hinders comparison and evidence-based policymaking.

4.3. Foresight and Emerging Trends Across Energy Community Dimensions

The rapid technological, regulatory, and social evolution of ECs requires analytical frameworks that move beyond current models centered on individual photovoltaic self-consumption. Recent scientific literature highlights a transition from instrumental perspectives—focused primarily on economic savings—toward transformative approaches aimed at a systemic reconfiguration of the energy sector. This section briefly discusses emerging aspects that should be incorporated into the EC monitoring framework, around the four dimensions analyzed in this article.

From a technological perspective, the frontier is shifting from simple energy generation toward digitalization and sector coupling. The future of ECs involves the integration of technologies such as Digital Twins for real-time optimization, Blockchain to enable peer-to-peer energy markets, and the management of Virtual Power Plants, which aggregate distributed energy resources to provide grid services. In parallel, a deeper integration of electric mobility is expected through Vehicle-to-Grid systems, alongside the incorporation of new energy vectors such as green hydrogen and collective thermal systems, enabling multidimensional energy autonomy (Lazaroïu, 2025).

In the governance dimension, future trends are shifting from a purely institutional focus to a conception of governance as an evolutionary and relational process that must adapt to the project's life cycle (Bauwens et al., 2022). In the Ibero-American context, this implies the formalization of a “hybrid governance” model that legally integrates informal and de facto governance structures (such as Community Action Boards or Indigenous self-governments) that already operate successfully in rural and peri-urban areas. The institutional maturity of these models will depend on the strengthening of Ostrom's principles, particularly with respect to community monitoring mechanisms (DP4) and graduated sanctions (DP5), which are areas identified as the main weaknesses in the region. Looking forward, governance must evolve to a polycentric and multilevel structure, where energy justice (distributive, procedural, and recognition) serves as the guiding principle to integrate households facing energy poverty through internal social tariffs. Finally, long-term sustainability will require institutionalizing gender equity and intergenerationality not as aspirational goals, but as constitutive operational strategies and succession plans that ensure leadership renewal and the persistence of community values against market pressures.

In the regulatory and financial spheres, future transitions are moving toward regional harmonization and bridging the structural regulatory divide identified between the Iberian Peninsula and Latin America. Legal frameworks will evolve from the restrictive promotion of individual self-generation toward robust figures of collective self-generation and community commercialization, enabling surplus compensation across multiple injection and consumption points. To ensure viability in energy poverty contexts, the future demands the creation of public guarantee funds and the design of direct subsidies for initial capital expenditures, preventing the transition from depending solely on members' capitalization capacity. Furthermore, ECs financing will migrate toward social and solidarity economy models, integrating green bonds, cooperative crowdfunding, and innovative schemes such as "Tax Credit Works".

This shift will reduce dependence on volatile international cooperation and shield financial autonomy against purely instrumental visions or potential capture by traditional commercial actors.

Finally, in the social appropriation and energy justice dimension, regional outlooks point toward the consolidation of restorative justice as an essential fourth pillar, aimed at remedying historical and territorial harms caused by previous extractive energy models in Indigenous and Afro-descendant communities. The future of ECs in Ibero-America requires transitioning from formal or consultative participation toward dignified energy appropriation, a process where technology is not imposed but co-constructed by the community to meet its priorities in health, education, and productive development (Bauwens et al., 2022). Such approach will integrate specific distributive justice mechanisms, as internal social tariffs and subsidized fees for households facing energy poverty, ensuring that local benefits prevail over financial profit. Furthermore, social sustainability will depend on closing the digital divide, guaranteeing community data sovereignty, and preventing the implementation of smart infrastructures (smart meters) from becoming a new barrier to exclusion. Finally, the justice dimension must institutionalize gender equity and intergenerationality as cross-cutting axes, ensuring not only parity in leadership but also the recognition of the differentiated energy needs of women and youth in marginalized rural and urban environments.

4.4. Toward an Ibero-American Observatory of Energy Communities

The identified gaps, and particularly the sixth, the absence of a standardized taxonomy, ground the proposal for an Ibero-American Observatory of Energy Communities as an open knowledge infrastructure. The Observatory is articulated around the four-dimensional taxonomy developed in this article, as it is shown in Figure 2, (Figure 3), (Figure 4) and (Figure 5). However, it is noted that this taxonomy does not treat these dimensions as isolated compartments but rather as an integrated system. In this framework, energy justice serves as the axiological core of values, and governance acts as the execution logic or operating system that facilitates the circulation of social and technical value. This system is physically enabled by the technological dimension, which provides the physical and digital environment for innovation and energy management, and is conditioned by the regulatory-financial dimension, which establishes the boundary conditions and stability of the regime required for the legal and economic viability of the community (Table 7).

The proposed taxonomy in this paper is intrinsically multidimensional, as terms across different dimensions are functionally interconnected; for example, achieving "liquid delegation" in governance (Figure 3) requires prior progress in "digital divide closure" and "energy literacy" within the social appropriation dimension (Figure 5). Furthermore, although not explicitly shown on the radial maps to preserve visual clarity, most indicators are designed to be assessed through maturity levels (Basic, Intermediate, and Advanced), allowing longitudinal tracking of the community's lifecycle. Finally, most of the terms in the proposed taxonomy form a glossary which serves as the formal coding framework for the RIPCEL Observatory, ensuring methodological consistency and data interoperability across empirical case studies in the Ibero-American region, see the appendix.

Table 7: Proposed Multidimensional Taxonomy and Indicators for the RIPCEL Observatory

Dimension	Category (Map Branch)	Suggested Indicators for the Observatory
1. Technological (Physical and Digital Substrate)	Generation and Storage Infrastructure	Installed capacity (kW) by energy source (Solar PV, Wind, Biomass, etc.). Storage capacity (kWh) and technology type (Li-Ion, H2, Thermal).
	Digital Management and Tech Frontier	Automation level (Basic EMS vs. AI-optimized management). Adoption of emerging technologies (Blockchain/P2P, VPP/V2G, Digital Twins).
	Cybersecurity and Data	Implementation of encryption protocols and access controls. Interoperability level of smart metering data.
2. Governance (Instrumental Framework)	Organizational Model and Maturity	Legal entity type (Cooperative, Association, Indigenous self-government, Informal). Compliance with Ostrom's Principles (specifically DP4 monitoring and DP5 sanctions).
	Power and Decision Dynamics	Voting mechanism (1p/1v vs. capital-weighted). Leadership rotation rate and existence of formal succession plans.
	Equity and Capacities	Percentage of gender parity and active youth/senior participation. Professionalization index (Voluntary vs. delegated management).
3. Regulatory-Financial (Boundary Conditions)	Legal Viability and Stability	Type of enabling framework (Collective self-generation vs. individual schemes). Regulatory stability index (perceived rate of normative change and predictability).
	Financing Mechanisms	Capital composition (% community equity, crowdfunding, public grants/subsidies). Use of innovative instruments (Tax Credit Works, Green Bonds, guarantee funds).
	Regime Relationship with DSO	Grid connection status and surplus compensation regime (Net-metering/Net-billing). Access to local flexibility markets and dispute resolution mechanisms.
4. Appropriation & Justice (Axiological Core)	Energy Justice (Distributive and Restorative)	% of bill savings for households facing energy poverty. Level of social reinvestment of surpluses into community infrastructure.
	Procedural and Recognition Justice	Level of technical co-construction (community participation in design). Alignment with territorial rights and use of local/ethnic languages.
	Agency and Community Sovereignty	Energy literacy index (internal training programs). Digital divide closure level and community data sovereignty.

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Table 7: Proposed Multidimensional Taxonomy and Indicators for the RIPCEL Observatory

Dimension	Category (Map Branch)	Suggested Indicators for the Observatory
5. Cross-cutting Dimension	Dissemination and Articulation	Science: Utilization of open repositories and shared datasets. Articulation: Integration with local organizations and regional networks.
	Framework Logic & Application	Interconnectivity: Analysis of cross-dimensional links (e.g., how digital closure enables liquid delegation). Tracking: Assignment of Basic-Advanced status to indicators for longitudinal evolution analysis.

Note: This multidimensional taxonomy serves as the primary coding framework for the RIPCEL Observatory. Most indicators are evaluated through maturity levels: Basic, Intermediate, and Advanced, to track the EC’s lifecycle.

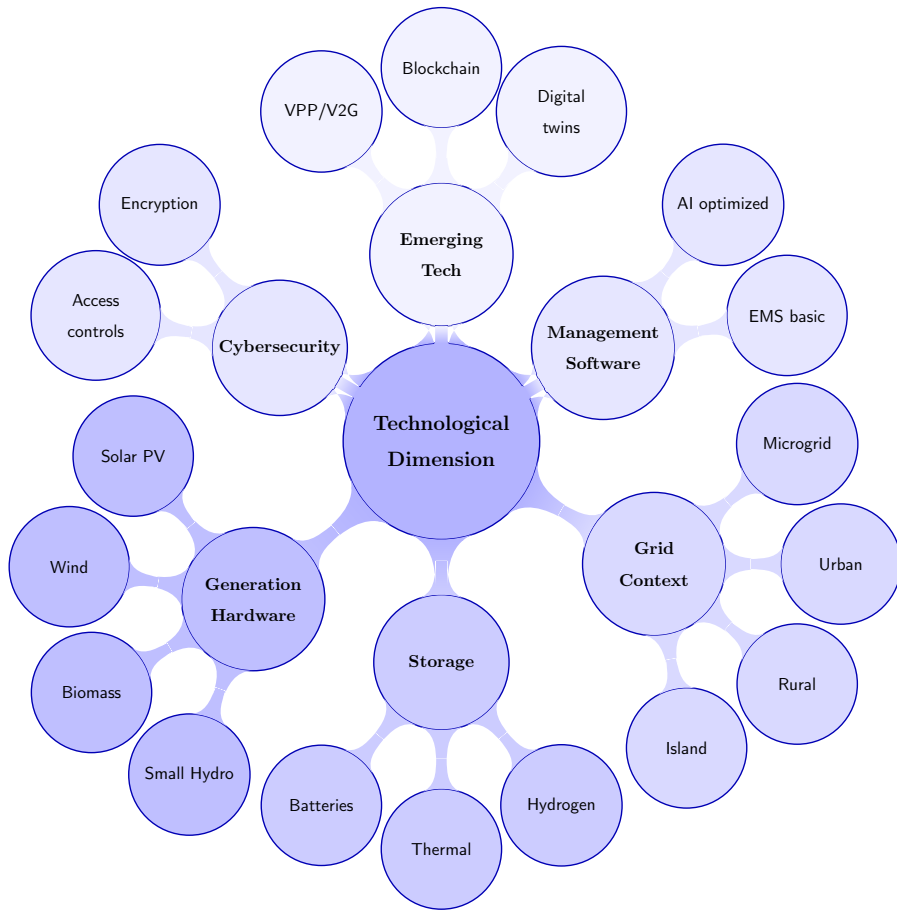


Figure 2: Technological dimension of CE taxonomy.

Proposed architecture and implementation feasibility. The Observatory is envisioned as a **federated** infrastructure, drawing lessons from three precedent observatories: the EU Energy Poverty Observatory (EPOV, now the Energy Poverty Advisory Hub), IRENA’s Renewable Energy Statistics program, and the Latin American Energy Outlook of OLADE. Key design decisions informed by these precedents:

Open repository Georeferenced database following the DCAT-AP metadata standard for interoperability, with each national RIPCEL node responsible for local data curation and the coordinating node (CYTED-funded for the initial

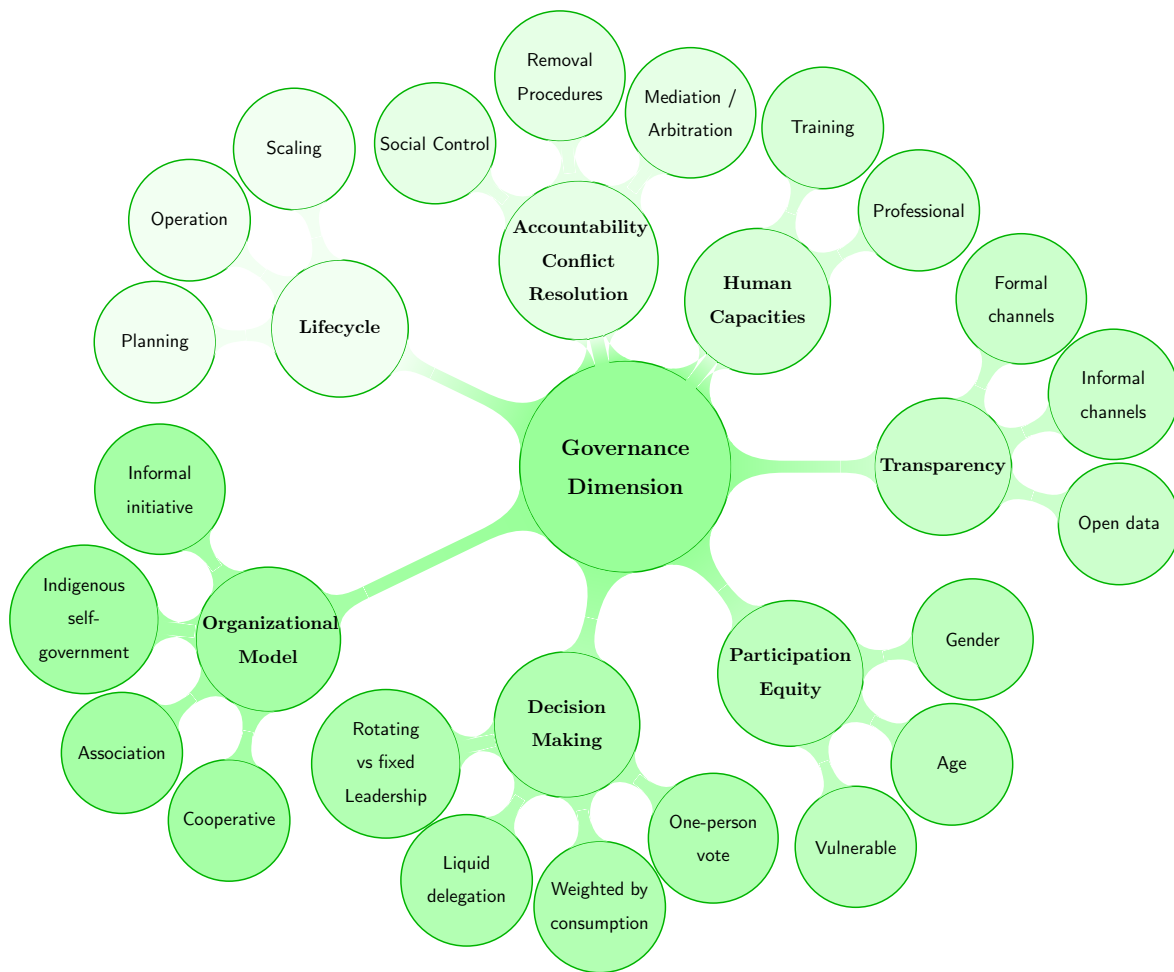


Figure 3: Governance dimension of CE taxonomy.



Figure 4: Regulatory-financial dimension of CE taxonomy.

4-year cycle) maintaining the central schema. Estimated annual hosting cost: USD 5,000–10,000 using open-source platforms (CKAN or Dataverse).

Monitoring network Standardized protocol of 25 technical, social, and governance indicators for longitudinal monitoring, aligned with SDG 7 reporting frameworks to facilitate integration with national statistical offices.

Knowledge platform Multilingual digital library (Spanish, Portuguese, English) with case studies, guides, tools, and training materials.

Community of practice Exchange forum for researchers, practitioners, policymakers, and community actors from the 15 RIPCEL countries.

Sustainability model The federated design distributes maintenance costs across national nodes, reducing dependence on a single funding source. Post-CYTED sustainability will be pursued through integration with the OLADE institutional framework and alignment with IDB’s energy transition lending programs.

Addressing survivorship bias To counteract the publication and survivorship bias identified in the literature (Section 4), the Observatory should adopt a registration model analogous to clinical trial registries: all EC initiatives would be registered at inception—not only upon achieving operational status—enabling longitudinal tracking of projects that are abandoned, stalled, or restructured, and generating evidence on failure modes that the current literature systematically omits.

4.5. Transferability beyond Ibero-America

While this review focuses explicitly on the Ibero-American context, the four-dimensional taxonomy and the Ostrom-based governance assessment developed here are inherently transferable across contexts. Energy community challenges in other

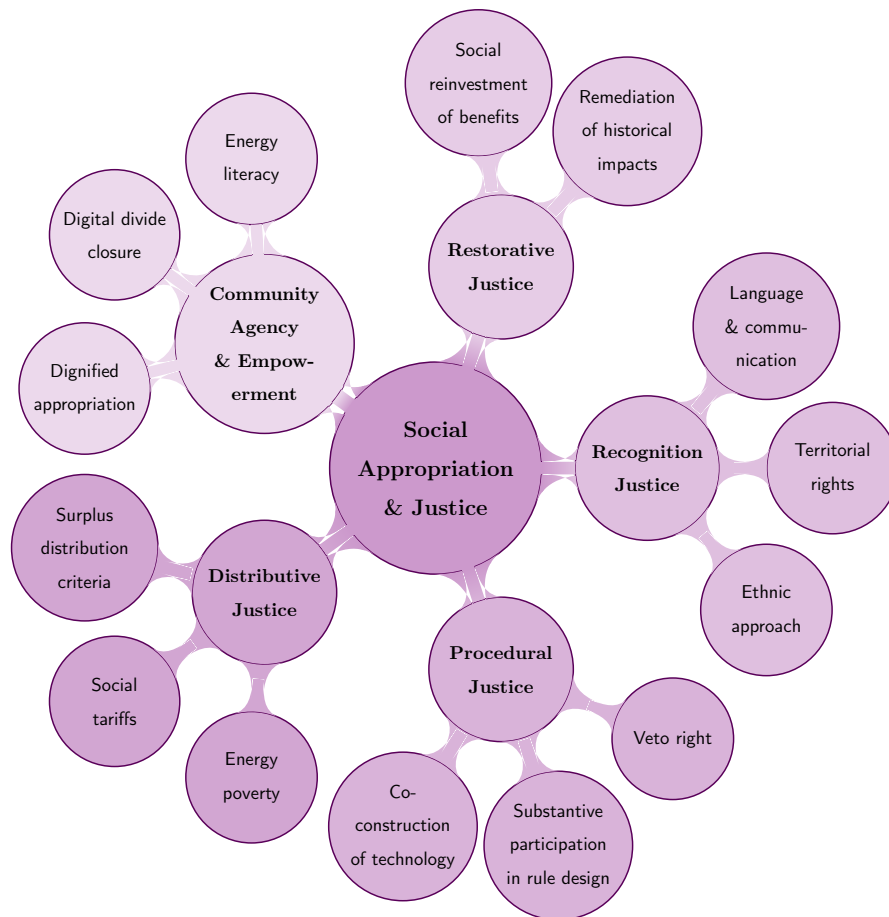


Figure 5: Social appropriation and justice dimension of CE taxonomy.

Global South regions—sub-Saharan Africa, South and Southeast Asia, Pacific Island States—share structural similarities with Latin America: colonial energy legacies, informal governance traditions, energy poverty, and the tension between centralized incumbent regimes and grassroots innovation. The proposed Observatory architecture, particularly the clinical-trial-registry model for tracking both successful and failed initiatives, could serve as a template for analogous regional observatories. Comparative research across these regions, using the taxonomy as a common analytical framework, would enable the identification of context-dependent versus universal enabling factors for energy community development.

5. Regulatory landscape of energy communities in Ibero-America: decentralization, energy justice, and energy system transformation

The Ibero-American region is at a critical juncture in which climate urgency converges with the imperatives of social justice and economic sovereignty. The energy transition in this geographic and cultural bloc has moved beyond a mere technological substitution of fossil fuels by renewables, becoming a profound restructuring of the relationships among the State, the market, and citizens (CEPAL, 2024; Ministerio de Minas y Energía de Colombia, 2024). Energy communities emerge as the principal vector of this transformation, enabling end-users to transcend their traditional role as passive consumers to become active agents of generation, management, and energy distribution.

5.1. The Iberian Peninsula: transposition and regional decentralization

Spain. The Spanish legal framework is based on Royal Decree-Law 23/2020, which partially transposes the EU Directives 2018/2001 (RED II) and 2019/944 (IEMD). This decree defines energy communities as legal entities based on open and voluntary participation, controlled by members who are natural persons, SMEs, or local authorities, with the primary objective of providing environmental, economic, or social benefits rather than financial profit (IDAE, 2024). Notably, while no new national law was enacted in 2024, the framework has been reinforced through ambitious autonomous community legislation: the Basque Country’s Law 1/2024 obliges that at least 20% of the capacity of new renewable projects be reserved for local citizen participation; Aragon’s Law 5/2024 regulates closed networks and promotes industrial self-consumption; and Galicia’s Law 2/2024 introduces mandatory social return obligations for projects using the territory’s natural resources (ECODES/EnergíaComún, 2024).

Portugal. Portugal’s trajectory has centered on collective self-consumption and renewable energy communities through Decree-Law 15/2022, which simplifies licensing procedures and allows energy surpluses to be shared more efficiently within communities, reducing dependence on the national grid and transport costs (VDA, 2022).

5.2. Latin America: emerging frameworks and institutional divergences

Colombia. Colombia has emerged as the Latin American leader in formalizing energy communities, linking them directly to its 2022–2026 National Development Plan (JET Knowledge, 2024). Decree 2236 of 2023 establishes two operational figures: the *Autogenerador Colectivo* (AGRC, for self-consumption with optional surplus injection) and the *Generador Distribuido Colectivo* (GDC, for full-grid injection up to 5 MW). Resolution CREG 101 072 of 2025 harmonizes these provisions with the national energy system, regulating commercial frontier registration, surplus accounting, and remuneration for injected energy (CREG, 2025). Despite the robustness of the legal framework, WRI and SEI research identifies financing and subnational knowledge gaps—particularly in the Pacific, Amazon, and Orinoco regions—as critical implementation barriers (WRI Colombia, 2025; Stockholm Environment Institute, 2025).

Mexico. Mexico’s 2025 energy reform, embodied in the *Ley del Sector Eléctrico* (LSE) and the *Ley de Planeación y Transición Energética* (LPTE), has elevated *Energy Justice* to a governing principle: any electricity project must now include

Free, Prior and Informed Consultation with indigenous and Afro-Mexican peoples, a Social Impact Statement (MIS), and a Social Management Plan (CEMDA, 2025; Holland & Knight, 2025). The threshold for the “Exempt Generator” classification (not requiring a full generation permit) was raised from 0.5 to 0.7 MW, facilitating small collective systems. However, the reform also centralizes planning under the *Secretaría de Energía* (SENER) and mandates that the *Comisión Federal de Electricidad* (CFE) maintain at least 54% of national generation, reconfiguring private and community space within a state-directed energy model.

Chile. Law 21.118 of 2018 extended Law 20.571 to enable energy generation for self-consumption in jointly owned properties and communities (Biblioteca del Congreso Nacional de Chile, 2018). Users connected to the same distribution network can register joint ownership of a generation asset and share surpluses via a Net Billing mechanism — a discount on the electricity bill. Energy cooperatives, which require at least five founding members, have emerged as the preferred legal vehicle and benefit from exemptions from contracted capacity payments (CNR, 2020).

Brazil. The *Marco Legal da Micro e Minigeração Distribuída* (Law 14.300, 2022) consolidated shared generation (*Geração Compartilhada*), allowing stakeholders to join consortia, cooperatives, or condominiums to install generating plants of up to 5 MW and offset participants’ consumption via a 60-month energy credit system (ANEEL, 2022). The regulation explicitly prohibits the splitting of generating plants to prevent regulatory arbitrage.

Table 8: Comparative regulatory features of energy communities across selected Ibero-American countries.

Country	Main ment	Instru- ment	Capacity Limit	Settlement Mechanism	Reference
Spain	RDL Aut. Laws	23/2020; Community	Varies (up to 100 MW)	Net billing; sur- plus trade	IDAE 2024; ECODES/EnergíaComún 2024
Portugal	DL 15/2022		1 MW (resi- dential)	Energy sharing credits	VDA 2022
Colombia	Decree 2236/2023; CREG 101 072/2025		<5 MW	Surplus injection tariff	JET Knowledge 2024; CREG 2025
Mexico	LSE + (2025)	LPTE	<0.7 MW (exempt)	Net metering (CFE)	CEMDA 2025; Holland & Knight 2025
Chile	Law 21.118 (2018)		300 kW (resi- dential)	Net billing dis- count	Biblioteca del Congreso Na- cional de Chile 2018; CNR 2020
Brazil	Law 14.300 (2022)		<5 MW	60-month energy credits	ANEEL 2022

5.3. Technical and institutional barriers

OLADE identifies 37 distinct barriers to the advancement of energy communities in Latin America and the Caribbean spanning political, regulatory, economic, technical, and social dimensions (OLADE, 2023). From a technical perspective, the massive integration of distributed energy resources (DER) creates voltage instability pressures in distribution networks;

the IDB reports that service interruptions in LAC exceed European averages by a factor of six to eight (IDB, 2022b). Smart metering is a prerequisite for viable surplus accounting and community management: without it, remuneration of community contributions is technically impossible or administratively prohibitive (IDB, 2019). Institutionally, vertically integrated utility models persist, limiting network access for community actors, with particular tensions in Bolivia and Colombia between small-scale generation promotion and the protection of state-owned incumbents (IDB, 2021).

5.4. Socioeconomic dimensions: gender, poverty, and territorial development

Energy community regulation in Ibero-America is intrinsically linked to reducing structural inequality. OLADE highlights that women in rural settings are disproportionately burdened by energy poverty, spending hours collecting biomass, and that energy communities offer a pathway to close intersectional gaps by integrating women in technical and governance roles (OLADE, 2024). From academia and cooperation agencies, the concept of *territorial social emancipation* has gained traction: energy should be understood not as a commercial commodity but as a democratically governed common good (OEI, 2025). Community co-design from project inception strengthens autonomy and social appropriation, reducing the social resistance often associated with large transnational renewable projects.

5.5. Strategic outlook: toward a modern regulatory ecosystem

Drawn from the evidence of CEPAL, the IDB, and OLADE, the following strategic directions underpin the policy recommendations in Section 6:

- (S1) *Strengthen binding national energy planning* aligned with local productive development and emissions reduction targets (CEPAL, 2024).
- (S2) *Modernize network remuneration frameworks* to recognize the stability, resilience, and equity value that communities contribute to the electricity system (IDB, 2022a).
- (S3) *Create innovative financing instruments*, guarantee funds, green bonds, cooperative crowdfunding, to reduce perceived risk in citizen-scale projects (KPMG, 2023).
- (S4) *Prioritize self-consumption in degraded urban areas and isolated rural communities*, preventing the occupation of high-value agricultural or ecological land (OEI, 2025).

The convergence of legal, financial, and social regulatory dimensions confirms the finding from Section 3: the regulatory divide between the Iberian Peninsula, enabled by EU directives and autonomous community innovation, and Latin America, building its own “energy justice” models, is the principal structural barrier to mass deployment of energy communities in Ibero-America.

6. Conclusion and policy implications

This systematic review has synthesized empirical evidence from 55 studies across 10 Ibero-American countries, providing, to the best of our knowledge, the first systematic review integrating the technological, governance, regulatory, and social appropriation dimensions of energy communities in the Ibero-American space under a unified analytical framework grounded in socio-technical transitions theory (MLP and TIS), governance of the commons, and the four pillars of energy justice.

Ibero-American ECs demonstrate significant potential for advancing just and sustainable energy transitions. Solar photovoltaic technology with storage and microgrids constitutes the most documented pathway, though not necessarily the most appropriate for all contexts, given the underrepresentation of alternatives in the literature. Cooperative models offer robust frameworks for democratic governance but require adaptation to local legal and cultural realities. The regulatory divide

between the Iberian Peninsula and Latin America emerges as the principal barrier to mass deployment. Energy justice must be central, not peripheral, to EC design, with the most marginalized communities as priority beneficiaries.

6.1. Policy implications

The following policy recommendations are derived from the synthesized evidence and designed to be actionable by policymakers, regulators, development banks, and community organizations across the region. This section is written to be comprehensible independently of the preceding technical analysis.

Regulatory frameworks.

- R1. *Adopt energy-community-specific legislation in Latin America.* Currently, most Latin American countries regulate only individual distributed generation, not collective community arrangements. Legislative frameworks should incorporate: (a) a legal definition of energy communities that requires participatory governance as a constitutive element (not merely shared generation); (b) flexible legal entities (cooperative, association, or indigenous community with self-governance); and (c) compensation regimes allowing communities to distribute surplus energy benefits among members. European directives cannot be directly transposed, as Latin American electricity markets feature vertically integrated state-owned utilities without an equivalent in liberalized European markets. Colombia’s Decree 2236/2023 provides an initial Latin American model that distinguishes self-generation ECs from commercialization ECs.
- R2. *Pursue regional regulatory harmonization.* Ibero-American cooperation mechanisms (OLADE, the Economic Commission for Latin America and the Caribbean, CYTED) should facilitate convergence toward a minimum operational energy community definition, following the taxonomy proposed in this article. This would enable the exchange of experience and economies of scale across the 15 RIPCEL countries.
- R3. *Simplify administrative procedures.* Grid connection, registration, and benefit distribution procedures must be simplified. Colombia’s single registry for energy communities and the Basque Country’s one-stop administrative window offer concrete models adaptable to local institutional capacity.

Financing.

- R4. *Design financing mechanisms for low-income communities.* Studies report financing gaps of USD 500–3,000 per capita for 5–50 kW community systems. Required measures include: (a) direct subsidies covering the majority of capital costs for communities below the multidimensional poverty line, with subsidy levels calibrated to national per-capita income and energy cost benchmarks; (b) concessional credits with grace periods aligned with project commissioning timelines; and (c) public guarantees mitigating perceived risk for commercial lenders. Battery storage, representing 30–50% of total system cost, requires dedicated investment instruments.
- R5. *Replicate innovative financial instruments.* Tax Credit Works (Colombia), Green Bonds (Costa Rica), and cooperative crowdfunding (Spain) offer documented models. Evidence from the reviewed studies suggests that blending public funds with cooperative capital and international cooperation produces the most sustainable financial structures.

Governance and capacity building.

- R6. *Develop flexible governance toolkits.* Following the institutional design principles of [Ostrom \(2009\)](#), modular guides should be developed enabling communities to choose among cooperative, associative, municipal, or mixed models suited to their legal and cultural context. These guides should include templates for: bylaws, benefit distribution regulations, collective decision protocols, and conflict resolution mechanisms.

- R7. *Invest in social capital and technical capacity.* Transition programs must prioritize training in participatory governance, financial management, and technical operations to prevent installed infrastructure from becoming non-functional due to a lack of community appropriation.
- R8. *Institutionalize university-community partnerships.* Universities must formalize their role as long-term energy community partners, moving beyond publication-centered academic incentives to value territorial impact through adjusted evaluation criteria.

Justice and social appropriation.

- R9. *Guarantee ethnic and gender differential approaches.* Policies must incorporate free, prior, and informed consultation mechanisms for indigenous peoples, and integrate explicit gender equity strategies into energy community governance. The fact that only 9% of reviewed studies address gender indicates a gap in both practice and research.

This article contributes to the literature by: (i) offering the first multilingual and four-dimensional systematic review of ECs in Ibero-America, complementing the analyses of [Polanco and García-Martínez \(2025\)](#) and [Peeters et al. \(2025\)](#); (ii) proposing a four-dimensional taxonomy anchored in MLP, TIS, governance of the commons, and energy justice frameworks, with the first operationalization of Ostrom’s principles against documented EC governance cases; (iii) identifying six critical knowledge gaps, confirming the optimistic bias noted in the literature; and (iv) laying the conceptual and operational foundations for an Ibero-American Observatory of Energy Communities with a feasibility-grounded architecture.

6.2. Limitations

This study presents the following limitations that should be considered when interpreting the results:

Database coverage. Although Scopus, Web of Science, SciELO, and RedALyC were incorporated to capture high-impact and regional literature, some grey literature (internal NGO reports, localized policy documents, local doctoral theses) may still be unrepresented. Future updates of this review should incorporate more unstructured sources.

Geographic bias. 33% of studies come from Spain, reflecting greater Iberian academic production but not necessarily a higher intensity of community experiences. The six unrepresented countries (Cuba, Ecuador, Honduras, Panama, Paraguay, Dominican Republic) constitute a significant gap.

Optimistic bias. Consistent with the warning of [Peeters et al. \(2025\)](#), none of the 55 included studies explicitly reports a failed or abandoned EC project. This suggests a publication bias that may overestimate success rates and underestimate the magnitude of structural barriers.

Cross-sectional design. Most studies are cross-sectional, precluding assessment of the long-term sustainability of documented ECs. Projects that appeared successful in their pilot phase may have faced subsequent challenges that were not captured.

Data extraction. Although inter-coder reliability metrics are robust (overall $\kappa = 0.85$, with dimension-specific values ranging from $\kappa = 0.78$ for social appropriation to $\kappa = 0.92$ for technology), manual extraction by human reviewers always introduces some degree of subjectivity in the categorization of ambiguous thematic dimensions such as restorative justice.

Proportions as descriptive frequencies. All percentages reported refer to the reviewed sample of 55 studies and should not be interpreted as population-level prevalence estimates. The small sample precludes formal statistical inference. Vote counts are reported following SWiM guidelines as a descriptive synthesis approach, not as effect size estimators.

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CRedit authorship contribution statement

Gerardo Blanco: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. RIPCEL Research Consortium: Validation, Resources, Data curation.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article. In the interest of transparency, it is noted that the RIPCEL Network, which funds this study, has as its mission the promotion of energy communities in Ibero-America. The authors have endeavored to maintain a balanced analytical perspective, documenting both the achievements and the limitations and gaps identified in the literature.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used AI-assisted language tools for language improvement and translation from the original Spanish manuscript into English. Additionally, Claude (co-work version) was utilized as a secondary independent reviewer during the initial title and abstract screening phase (Stage 2) of the systematic review. After using these tools, the authors reviewed and edited all content as needed, performed additional screening of full conclusions in cases of human-AI discrepancy, and take full responsibility for the content of the published article.

Data availability

The extraction database containing records for the 55 included studies, the complete list of excluded studies with reasons for exclusion, the detailed search protocol, individual MMAT quality assessments (Supplementary Table S1), and manual analysis scripts are deposited in the RIPCEL project OSF repository (<https://osf.io/ripcel>). A supplementary file accompanying this manuscript contains Table S1 (MMAT scores).

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Appendix A. Glossary of terms for the proposed CE taxonomy

A. Technological dimension

Term	Definition
Solar PV	Photovoltaic panels that convert sunlight directly into electricity; the most common generation technology in CEs (89% of cases in the region).
Wind	Small-scale wind turbines used for local electricity generation, often complementary to solar PV.
Biomass	Organic materials (agricultural residues, wood, etc.) used for heat or power generation in rural CEs.
Small Hydro	Mini or micro hydropower plants (typically <10 MW) that exploit local water flows without large reservoirs.
Green H ₂	Hydrogen produced from renewable electricity via electrolysis; used for long-term storage or hard-to-electrify sectors.
Batteries	Electrochemical storage devices (Li-ion, lead-acid, etc.) that provide short- to medium-term flexibility.
Thermal storage	Systems that store heat (e.g., molten salts, hot water) for later use in heating or power generation.
Hydrogen storage	Storage of chemical energy as hydrogen gas, suitable for seasonal or multi-day backup.
Microgrid	A local, self-controlled grid that can operate connected to the main grid or in island mode.
Urban grid context	CEs located in cities, typically connected to a strong distribution network.
Rural grid context	CEs in low-density areas, often with weak or unreliable grid connection.
Island grid context	Isolated systems (geographic or technical) requiring high self-sufficiency.
EMS basic	Energy Management System with monitoring and basic control functions.
AI optimised	Advanced EMS using artificial intelligence for predictive control, load forecasting, and real-time optimisation.
Digital twins	Virtual replicas of the physical energy system used for simulation, testing, and training.
Blockchain	Distributed ledger technology enabling peer-to-peer energy trading, automated smart contracts, and verifiable voting.
VPP / V2G	Virtual Power Plant (aggregation of distributed resources) and Vehicle-to-Grid (use of EV batteries for grid services).
Encryption	Cryptographic protection of communication and data within the CE's digital infrastructure.
Access controls	Policies and mechanisms that define who can view or modify data and control parameters.
Cybersecurity	Overall protection of the CE's digital assets against unauthorised access, manipulation, or disruption.

B. Governance dimension

Term	Definition
Cooperative	A formal legal form where members own and democratically control the CE under the one-member-one-vote principle.
Association	A non-profit legal entity common for community initiatives; less strict than a cooperative but may have limitations for energy commercialisation.

Continues on next page

Term	Definition
Indigenous self-government	Governance structures based on traditional authorities and customary norms, recognised within the CE.
Informal initiative	A CE operating without formal legal recognition, often based on social trust and de-facto agreements.
One-person-one-vote	The basic democratic rule where every member has equal voting power regardless of investment or consumption.
Weighted by consumption	Voting power proportional to annual energy consumption (or investment). Less egalitarian but may reflect stake.
Liquid delegation	A hybrid system where members can vote directly on issues or delegate their vote to experts/leaders, revocable at any time.
Rotating vs fixed leadership	Whether board positions change periodically (rotating) or remain with the same individuals (fixed).
Succession plan	Documented procedure for replacing board members or key technical roles to ensure institutional continuity.
Gender equity	Measures to ensure balanced participation of women and men, e.g., board quotas, anti-harassment protocols.
Age inclusion	Specific provisions for youth and elderly representation in decision-making bodies.
Vulnerability inclusion	Mechanisms to include energy-poor households or socio-economically disadvantaged members.
Formal channels	Official communication means: recorded assemblies, minutes, official letters.
Informal channels	Daily communication via chat groups, social media, newsletters, or word-of-mouth.
Open data policy	Commitment to make generation, consumption, and financial data accessible to all members (respecting privacy).
Training	Formal education for members on technical, administrative, and democratic skills.
Professional capacities	Degree to which the CE hires external specialists vs. relying on voluntary work.
Peer review	Internal oversight where members audit each other's compliance with rules.
Recall procedures	Clear rules and thresholds for removing board members before their term ends.
Mediation / Arbitration	Internal conflict resolution mechanisms before resorting to courts.
Planning phase	Early stage of the CE where the founding assembly defines vision, rules, and initial investments.
Operation phase	Day-to-day management, usually delegated to a board or hired staff, with member oversight.
Scaling	Process of adding new members or expanding the CE's geographical scope without undermining democratic quality.

C. Regulatory-Financial dimension

Term	Definition
Enabling legal framework	The set of laws and regulations that explicitly recognise CEs, allow collective self-generation, and permit community surplus commercialisation.
Self-generation (collective)	Legal permission for a group of consumers to jointly own a generation facility and allocate the produced energy among themselves.

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Term	Definition
Community surplus commercialisation	The right of a CE to sell excess electricity to third parties or to the grid.
Regional harmonisation	Alignment of regulations across neighbouring jurisdictions (e.g., Iberian-Latin American context) to reduce cross-border barriers.
Crowdfunding	Small contributions from many individuals (often via online platforms) to finance CE projects.
Green bonds	Debt instruments issued by the CE or an intermediary, where proceeds are earmarked for renewable energy or energy efficiency.
Subsidies	Direct public financial support for initial investment, technical assistance, or reduced interest rates.
Private / impact investment	Capital from private funds or social investors that seek both financial return and measurable environmental/social impact.
Net billing / Net metering	Mechanisms for compensating the CE for electricity injected into the grid (the difference between purchased and injected energy).
Flexibility markets	Markets where CEs can sell grid services (demand response, voltage support, frequency regulation) to the distribution system operator (DSO).
Dispute resolution mechanisms	Procedures (administrative or judicial) to settle disagreements between the CE and the DSO or regulator.
Rate of regulatory change	Indicator of how frequently laws or rules affecting CEs are modified; high rates increase uncertainty.
Public-community dialogue	Institutionalised forums where CE representatives and public authorities discuss regulatory improvements.

D. Social Appropriation & Justice dimension

Term	Definition
Energy poverty	Household situation where a significant share of income is spent on energy, or basic energy needs are not met.
Social tariffs	Internal discounted electricity rates for low-income or vulnerable members, funded by cross-subsidies from other members.
Surplus distribution criteria	Rules for allocating financial surpluses from energy sales: proportional to consumption, equal share, or need-based.
Co-construction of technology	Participatory design process where community members decide which technologies (e.g., solar vs wind) and where to install them.
Substantive participation in rule design	Not just being consulted, but having binding power in creating the CE's internal rules.
Veto right	Power of certain groups (e.g., indigenous assembly, vulnerable members) to block decisions that affect them disproportionately.
Ethnic approach	Recognition of indigenous or traditional authorities as legitimate governance actors within the CE.
Territorial rights	Respect for prior consultation and self-determination when the CE operates on ancestral lands.

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Term	Definition
Language & communication	Use of native languages, accessible formats (large print, audio), and culturally appropriate channels.
Remediation of historical impacts	Compensatory actions for past harms caused by energy projects (e.g., displacement, pollution).
Social reinvestment	Channelling part of CE's profits into community health, education, or infrastructure funds.
Energy literacy	Understanding of basic electricity concepts, billing, self-consumption, and market options; enables informed participation.
Digital divide closure	Ensuring all members have affordable devices, connectivity, and digital skills to use the CE's platforms and data.
Dignified appropriation	Community's ability to modify, repair, and culturally adapt technology; opposes technological dependency and "black-boxed" systems.
Community agency & empowerment	The collective capacity to set priorities, control resources, and act autonomously in the energy field.